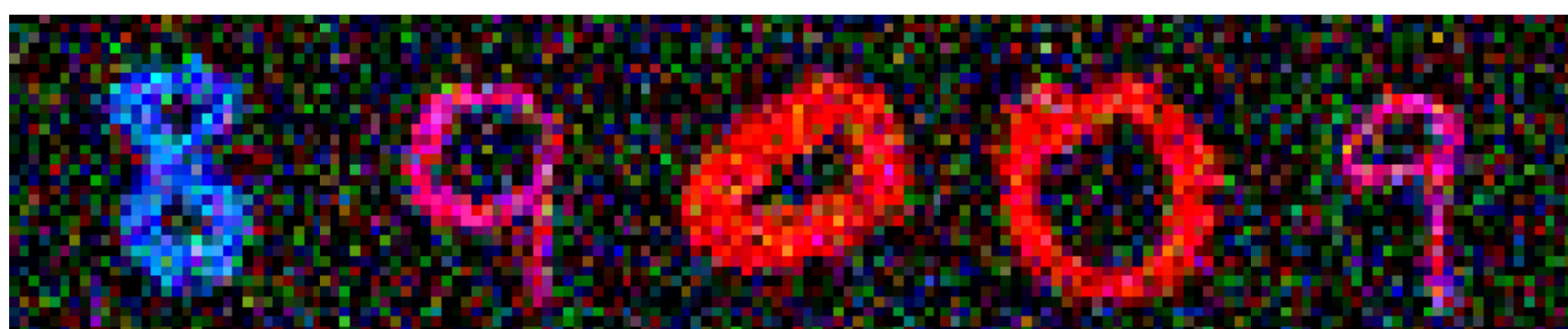
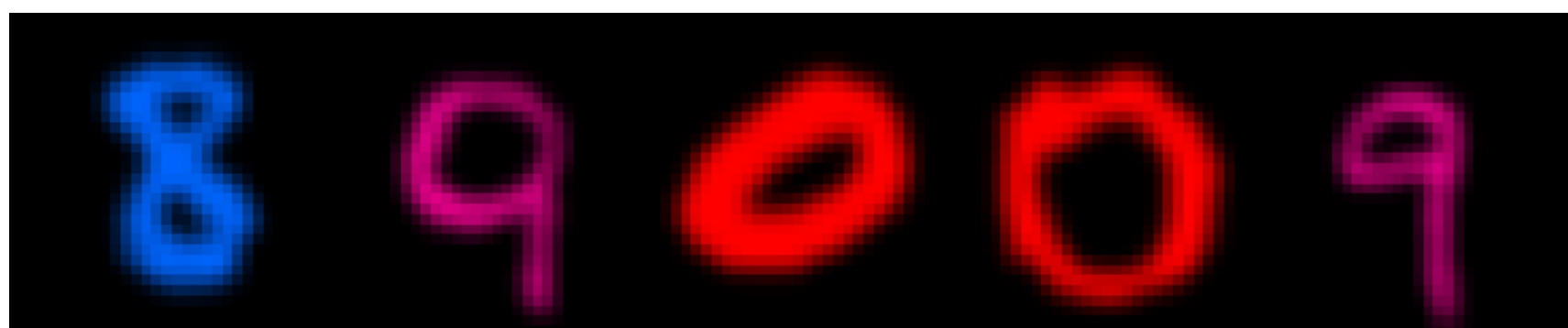
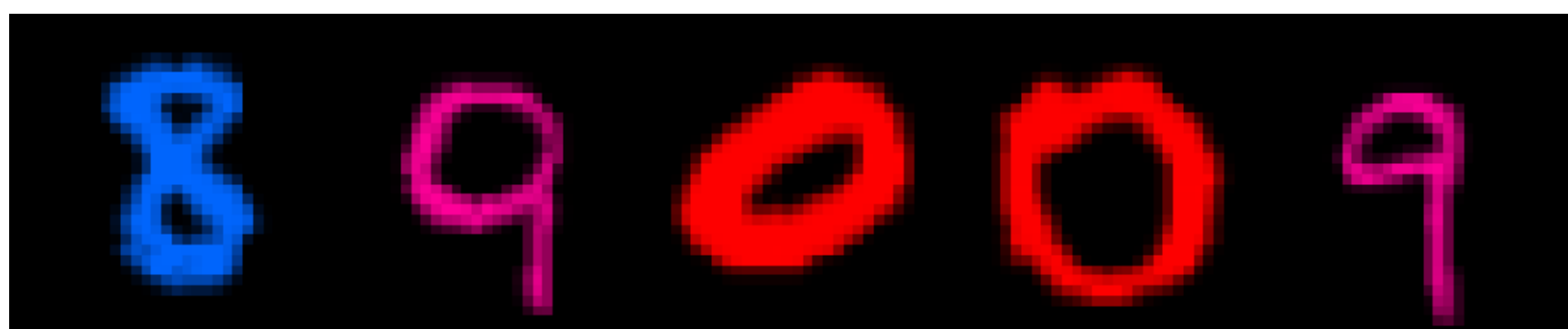
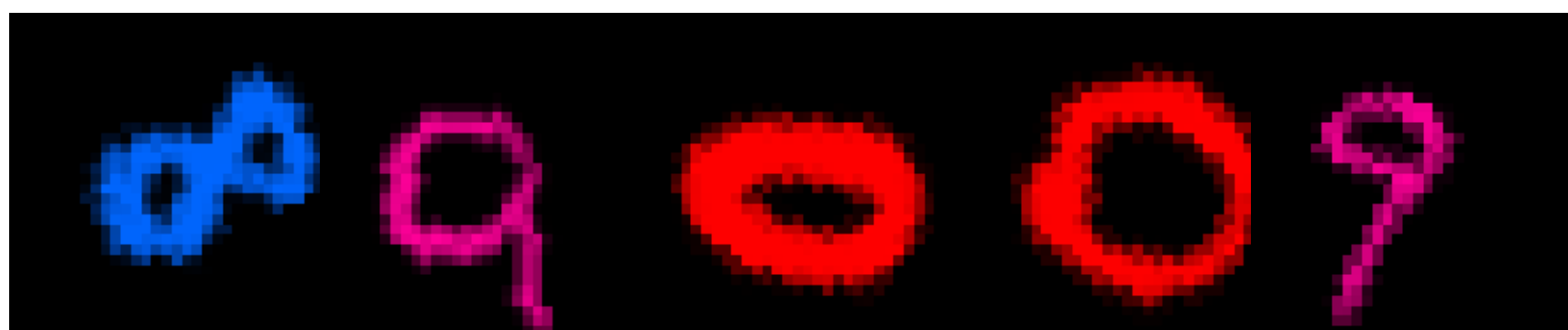
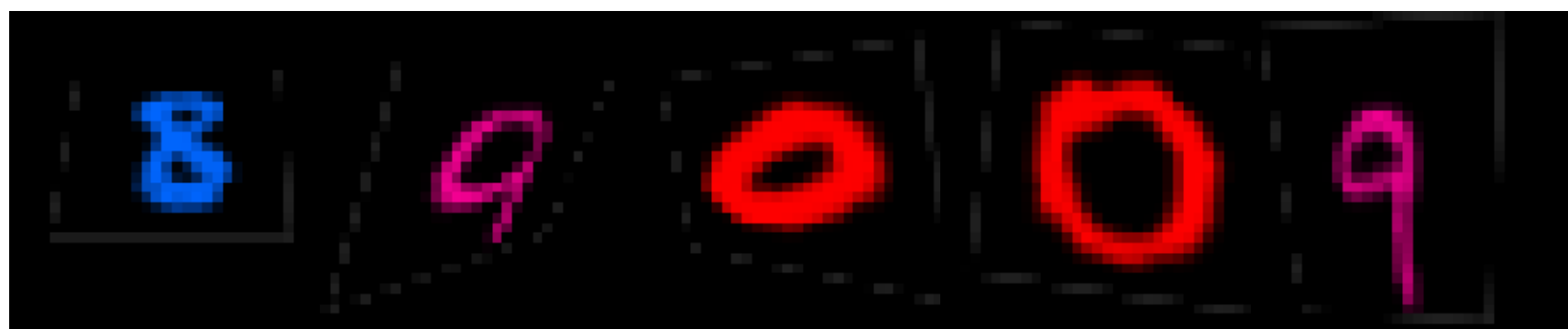
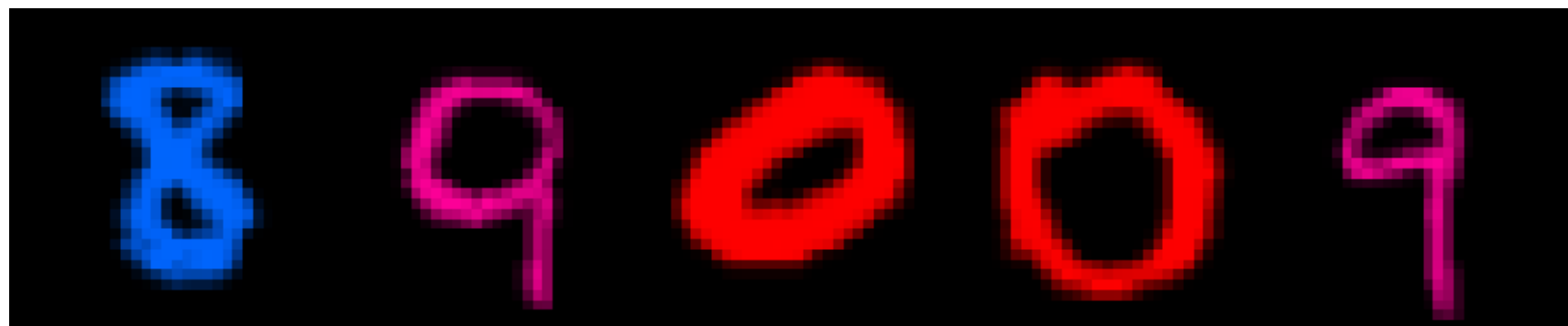
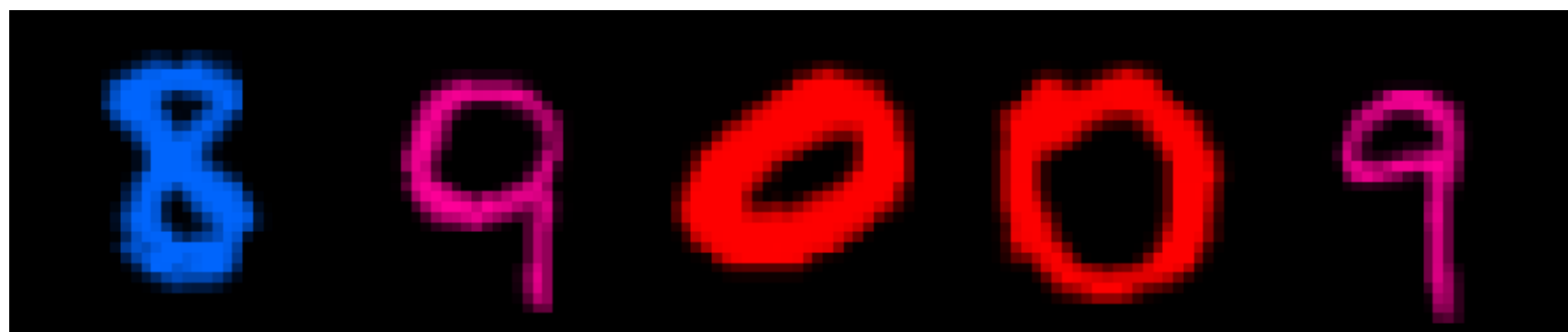


CMNIST Shifts: Camera + Huber (Visuals)



Camera Shift: Angle (Affine Corrected)

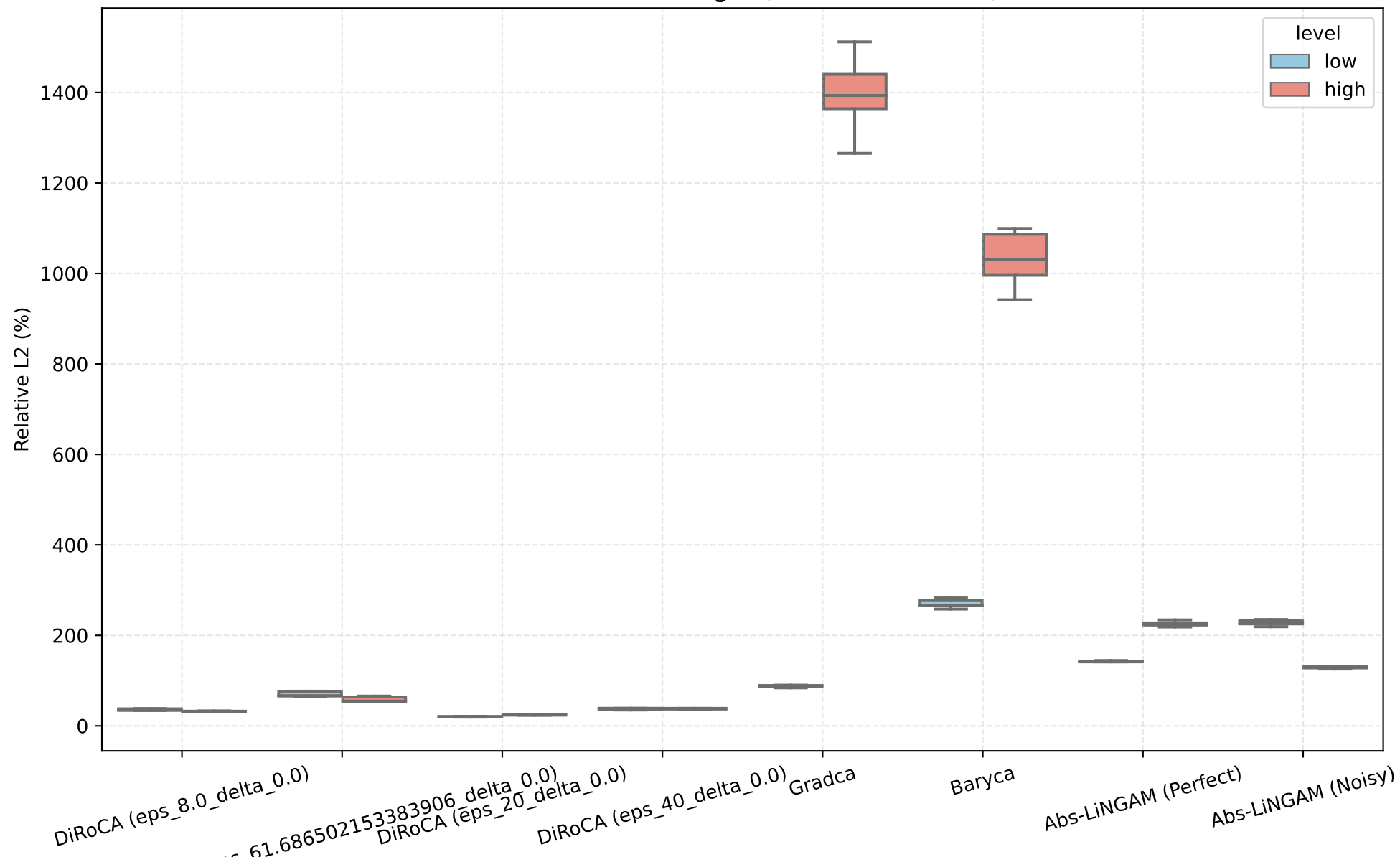


Table: Angle

| DiRoCA (eps_61.686502153383906_delta_0.0) | Method                                    | Low           | High            |
|-------------------------------------------|-------------------------------------------|---------------|-----------------|
|                                           | Abs-LiNGAM (Noisy)                        | 227.89 ± 6.09 | 130.58 ± 3.99   |
|                                           | Abs-LiNGAM (Perfect)                      | 141.63 ± 2.57 | 225.41 ± 4.40   |
|                                           | Baryca                                    | 270.19 ± 9.15 | 1032.72 ± 57.07 |
|                                           | DiRoCA (eps_20_delta_0.0)                 | 20.05 ± 0.73  | 24.02 ± 0.33    |
|                                           | DiRoCA (eps_40_delta_0.0)                 | 38.67 ± 3.37  | 38.66 ± 2.09    |
|                                           | DiRoCA (eps_61.686502153383906_delta_0.0) | 69.80 ± 5.30  | 58.43 ± 5.48    |
|                                           | DiRoCA (eps_8.0_delta_0.0)                | 35.97 ± 1.79  | 32.20 ± 0.65    |
|                                           | Gradca                                    | 87.64 ± 2.29  | 1387.58 ± 91.31 |

Camera Shift: Rotation (Affine Corrected)

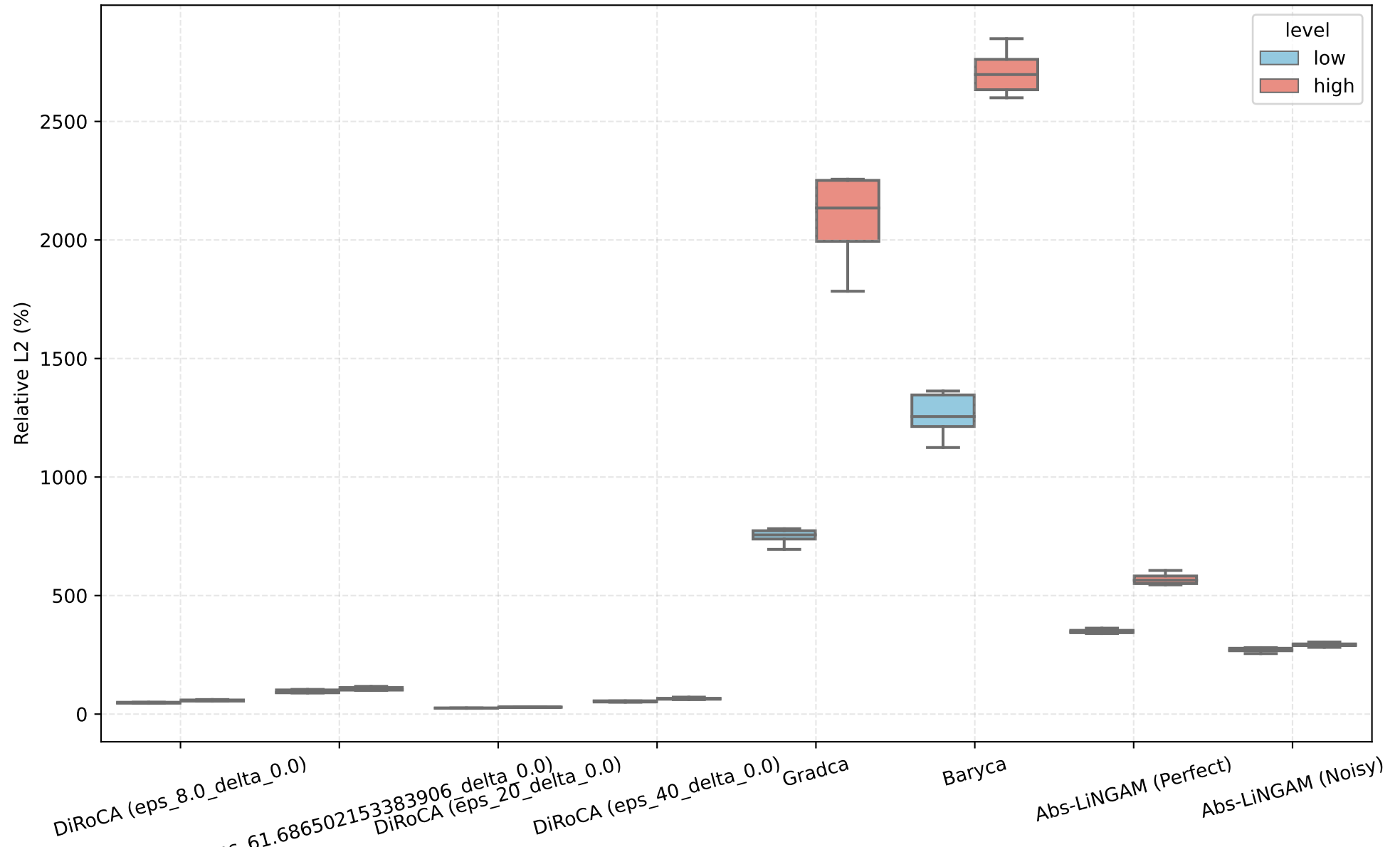


Table: Rotation

| DiRoCA (eps_61.686502153383906_delta_0.0) | Method                                    | Low             | High             |
|-------------------------------------------|-------------------------------------------|-----------------|------------------|
|                                           | Abs-LiNGAM (Noisy)                        | 270.15 ± 8.74   | 292.55 ± 7.65    |
|                                           | Abs-LiNGAM (Perfect)                      | 348.10 ± 10.43  | 568.41 ± 20.90   |
|                                           | Baryca                                    | 1260.26 ± 89.34 | 2709.12 ± 162.23 |
|                                           | DiRoCA (eps_20_delta_0.0)                 | 25.27 ± 0.53    | 29.21 ± 0.37     |
|                                           | DiRoCA (eps_40_delta_0.0)                 | 55.08 ± 4.74    | 65.42 ± 3.56     |
|                                           | DiRoCA (eps_61.686502153383906_delta_0.0) | 95.35 ± 6.46    | 106.72 ± 6.76    |
|                                           | DiRoCA (eps_8.0_delta_0.0)                | 47.90 ± 1.78    | 57.09 ± 2.59     |
|                                           | Gradca                                    | 752.25 ± 26.67  | 2197.35 ± 311.27 |

Camera Shift: Blur (Affine Corrected)

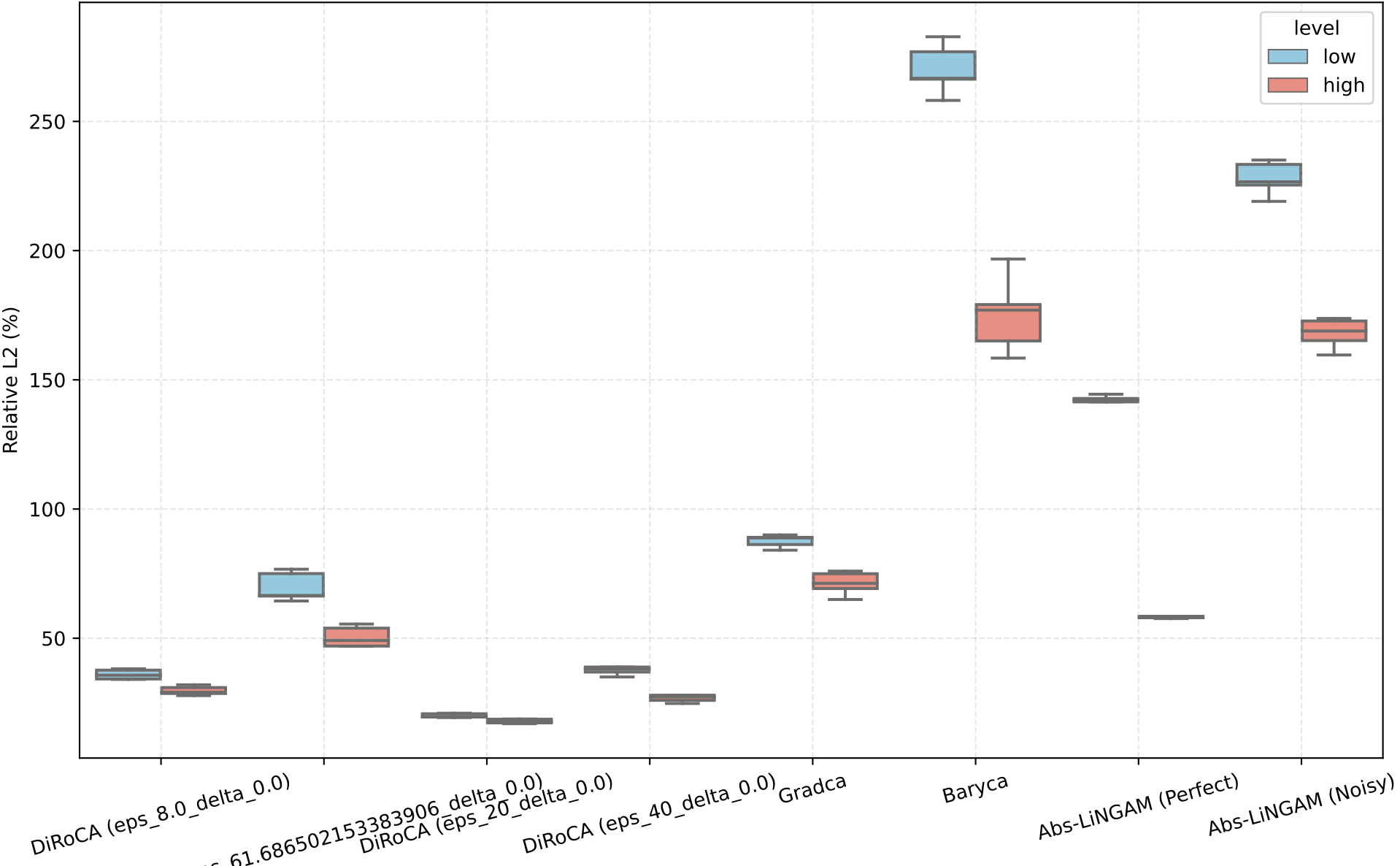


Table: Blur

| DiRoCA (eps_61.686502153383906_delta_0.0) | Method                                    | Low           | High           |
|-------------------------------------------|-------------------------------------------|---------------|----------------|
|                                           | Abs-LiNGAM (Noisy)                        | 227.89 ± 6.09 | 168.04 ± 5.48  |
|                                           | Abs-LiNGAM (Perfect)                      | 141.64 ± 2.57 | 58.92 ± 1.90   |
|                                           | Baryca                                    | 270.19 ± 9.15 | 175.27 ± 13.89 |
|                                           | DiRoCA (eps_20_delta_0.0)                 | 20.05 ± 0.73  | 17.84 ± 0.78   |
|                                           | DiRoCA (eps_40_delta_0.0)                 | 38.68 ± 3.37  | 27.75 ± 2.91   |
|                                           | DiRoCA (eps_61.686502153383906_delta_0.0) | 69.80 ± 5.30  | 50.48 ± 3.77   |
|                                           | DiRoCA (eps_8.0_delta_0.0)                | 35.97 ± 1.79  | 29.68 ± 1.62   |
|                                           | Gradca                                    | 87.64 ± 2.29  | 71.28 ± 4.21   |

Huber Contamination (Affine Corrected)

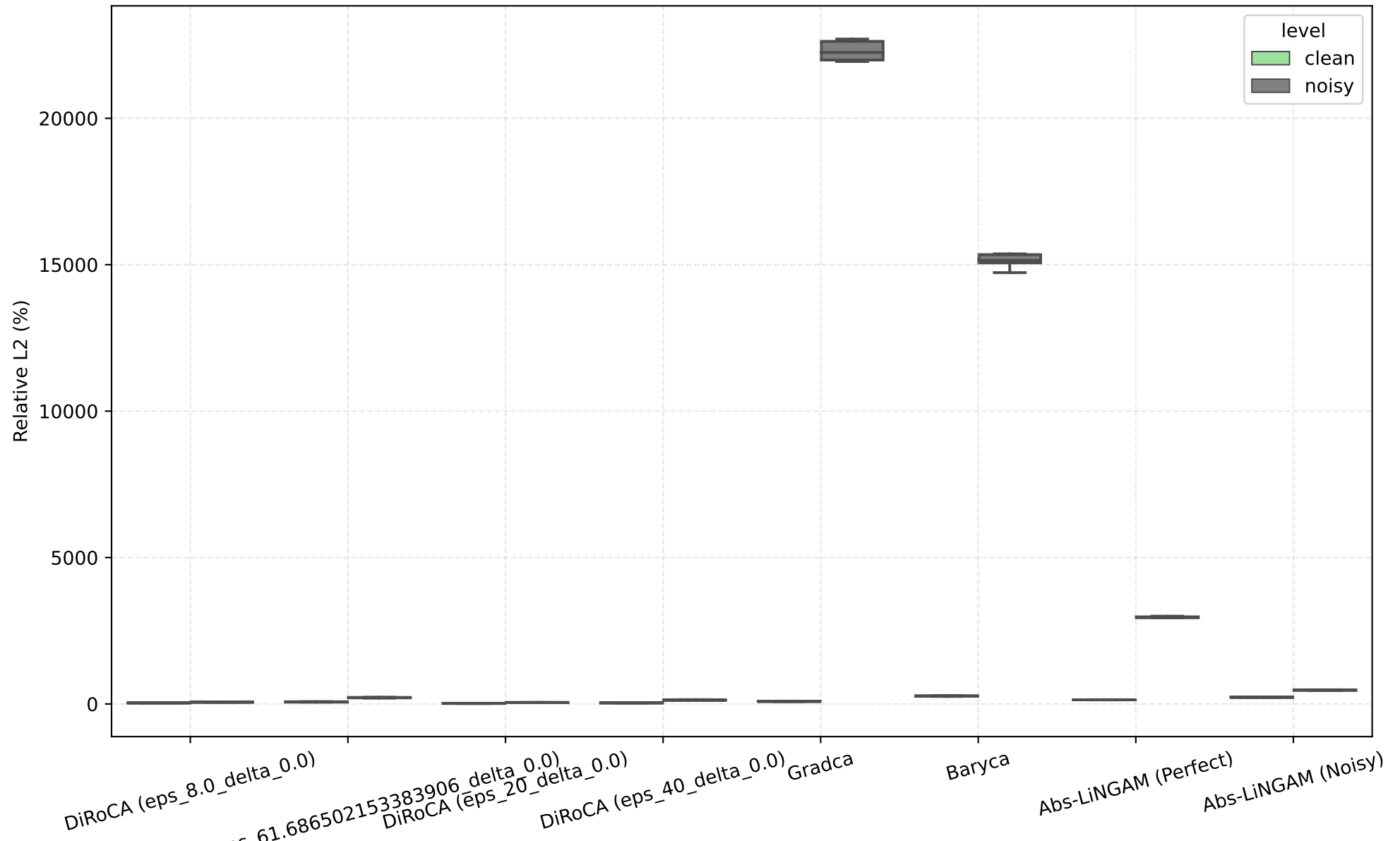




Table: Huber Contamination

| DiRoCA (eps_61.686502153383906_delta_0.0) | Method                                    | Clean ( $\alpha=0$ ) | Noisy ( $\alpha=1$ )   |
|-------------------------------------------|-------------------------------------------|----------------------|------------------------|
|                                           | Abs-LiNGAM (Noisy)                        | 227.89 $\pm$ 6.09    | 472.73 $\pm$ 8.90      |
|                                           | Abs-LiNGAM (Perfect)                      | 141.64 $\pm$ 2.57    | 2943.56 $\pm$ 57.25    |
|                                           | Baryca                                    | 270.19 $\pm$ 9.15    | 15312.06 $\pm$ 628.37  |
|                                           | DiRoCA (eps_20_delta_0.0)                 | 20.05 $\pm$ 0.73     | 48.21 $\pm$ 0.65       |
|                                           | DiRoCA (eps_40_delta_0.0)                 | 38.68 $\pm$ 3.37     | 132.58 $\pm$ 7.90      |
|                                           | DiRoCA (eps_61.686502153383906_delta_0.0) | 69.80 $\pm$ 5.30     | 215.22 $\pm$ 9.19      |
|                                           | DiRoCA (eps_8.0_delta_0.0)                | 35.97 $\pm$ 1.79     | 60.42 $\pm$ 1.73       |
|                                           | Gradca                                    | 87.64 $\pm$ 2.29     | 22266.84 $\pm$ 1542.40 |